



Batterjee Medical College
For Science and Technology

Lung Physiology primer & diagnostic studies

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Learning Objectives

Knowledge:

- Identify the anatomical divisions of the respiratory tracts, including lung lobar structures, the mediastinum, and pleural coverings.
- Differentiate between the conducting zone (anatomical dead space) and the respiratory zone.
- Compare the structure and physiological functions of Type I and Type II pneumocytes in alveolar gas exchange and surfactant production.
- Explain the mechanics of ventilation-perfusion (V/Q) matching, detailing the pathophysiological consequences of intrapulmonary shunting ($V = 0$) and dead-space ventilation ($Q = 0$).
- Define standard static and dynamic lung volumes and capacities (TV , IRV , ERV , RV , VC , FRC , and TLC) and state their normal physiological ranges.
- Distinguish between obstructive and restrictive lung diseases using spirometric indices, lung volumes, and flow-volume loops.
- Explain the indications and clinical utility of advanced lung function studies, including body plethysmography, DL_{CO} , and bronchoprovocation challenge testing.

Skills:

- Correlate physical examination findings with specific respiratory pathologies.
- Select appropriate diagnostic imaging and invasive modalities based on the suspected clinical presentation.
- Apply a systematic diagnostic algorithm to synthesize clinical findings, PFT results, and imaging to formulate a differential diagnosis.

Value:

- Demonstrate a professional attitude in adherence to the training site's rules, regulations, and clinical guidelines.



- Introduction
- Anatomy
- Physiological lung volumes
- Evaluation of the lung function
- Evaluation of the lung disease

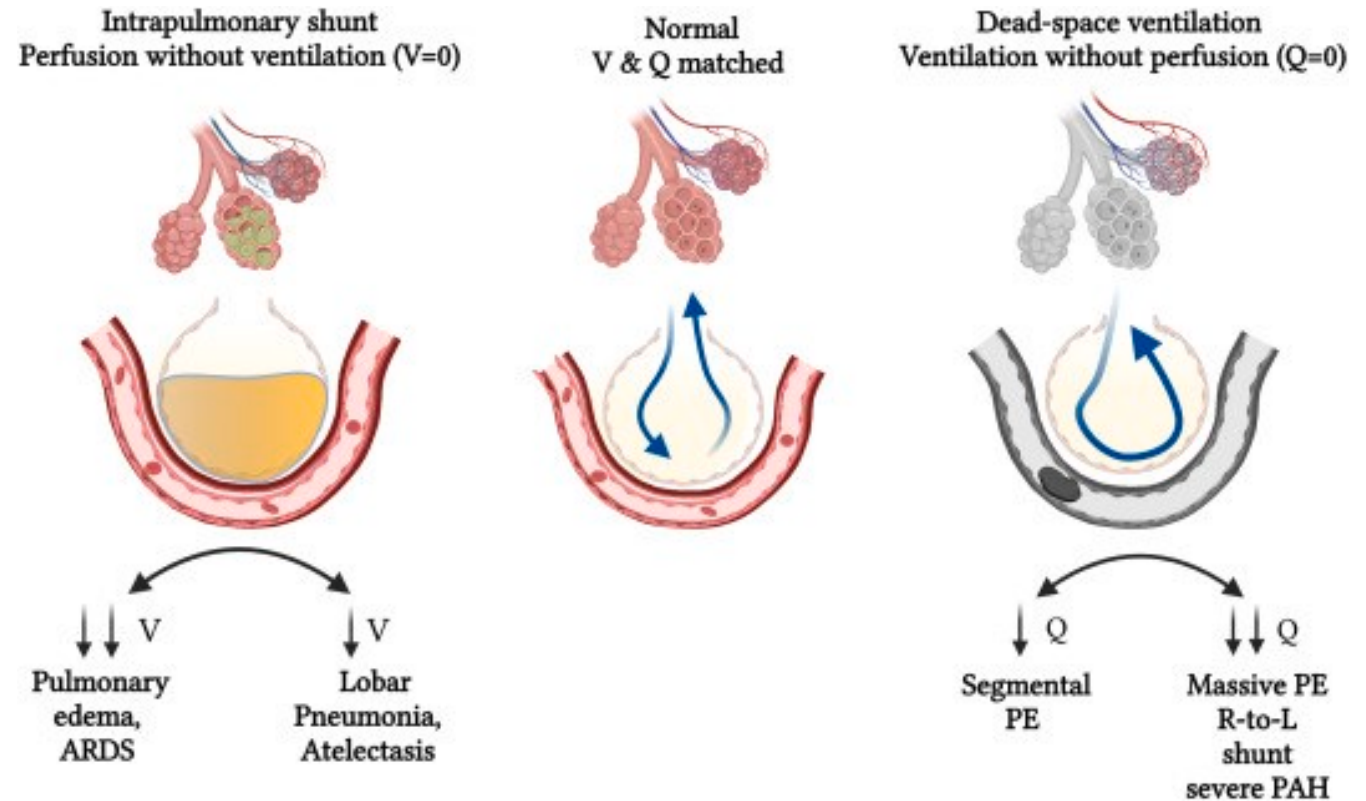
Introduction

Lungs Primary Function:

- Deliver oxygen to the pulmonary capillary blood and to excrete carbon dioxide

This occurs via:

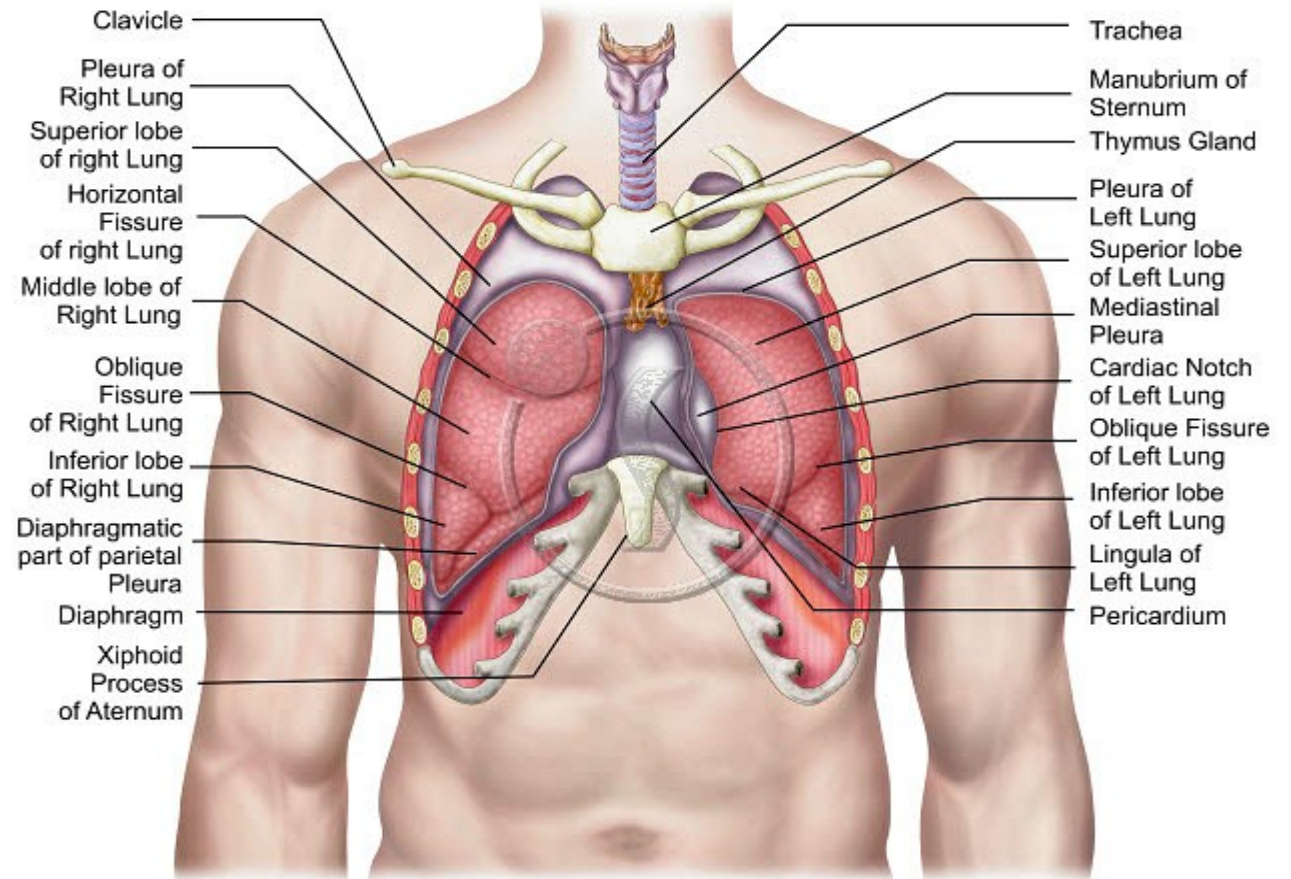
- Ventilation (V)
- Gas Exchange (Q)
- Ventilation-Perfusion matching



Anatomy

The lungs: major organs of the **respiratory system** are divided into sections, or lobes. The right **lung** has three lobes and is slightly larger than the left lung, which has two lobes.

The lungs are separated by the **mediastinum**. This area contains the **heart**, **trachea**, **esophagus**, and many **lymph** nodes. The lungs are covered by a protective **membrane** known as the **pleura** and are separated from the **abdominal cavity** by the muscular diaphragm.



Anatomy

With each **inhalation**, air is pulled through the trachea and the branching passageways of the lungs (the **bronchi**), filling thousands of tiny **alveoli** at the ends of the bronchi.

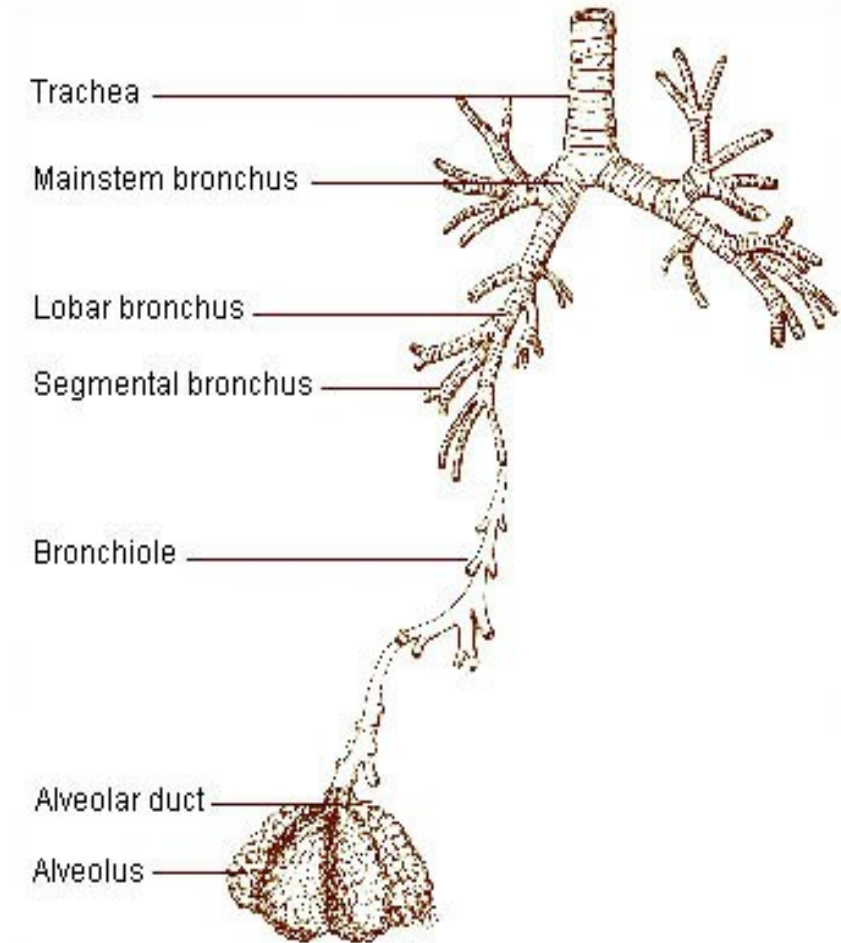
These sacs, are surrounded by small **blood** vessels **capillaries**. **Oxygen** passes through the thin membranes of the alveoli into the bloodstream.

The red blood cells pick up the oxygen and carry it to the body's organs and tissues. As the blood cells release the **oxygen** and they pick up **carbon dioxide**.

The carbon dioxide is then carried back to the lungs and released into the alveoli. With each **exhalation**, carbon dioxide is expelled from the bronchi out through the trachea.

Anatomy

The **respiratory tract** is divided into two main parts: the upper respiratory tract, consisting of the **nose**, **nasal cavity** and the **pharynx**; and the lower respiratory tract, consisting of the **larynx**, **trachea**, **bronchi** and the lungs. The trachea, which begins at the edge of the larynx, divides into two bronchi and continues into the lungs. The trachea allows air to pass from the larynx to the bronchi and then to the lungs. The bronchi divide into smaller **bronchioles** which branch in the lungs forming passageways for air. The terminal parts of the bronchi are the **alveoli**. The alveoli are the functional units of the lungs and they form the **site** of gaseous exchange.



Anatomy

Airway:

- The conducting zone (Anatomical dead space)
- The Respiratory zone

Airway subdivision	Order No.	Cross-sectional area (cm ²)	Resistance (cm H ₂ O • L ⁻¹ • sec)
Larynx	0		0.5
Trachea	0	2.5	0.5
Bronchi	1	2.0	
	2		0.2
Bronchioles	16	5.0	
	17	1.8 x 10 ²	
Respiratory bronchioles	19	9.4 x 10 ²	
Alveolar ducts	22	5.8 x 10 ³	
Alveoli	23	5.6 x 10 ⁷	

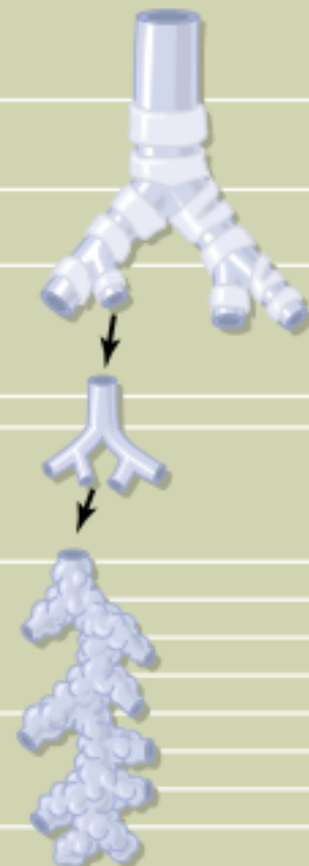
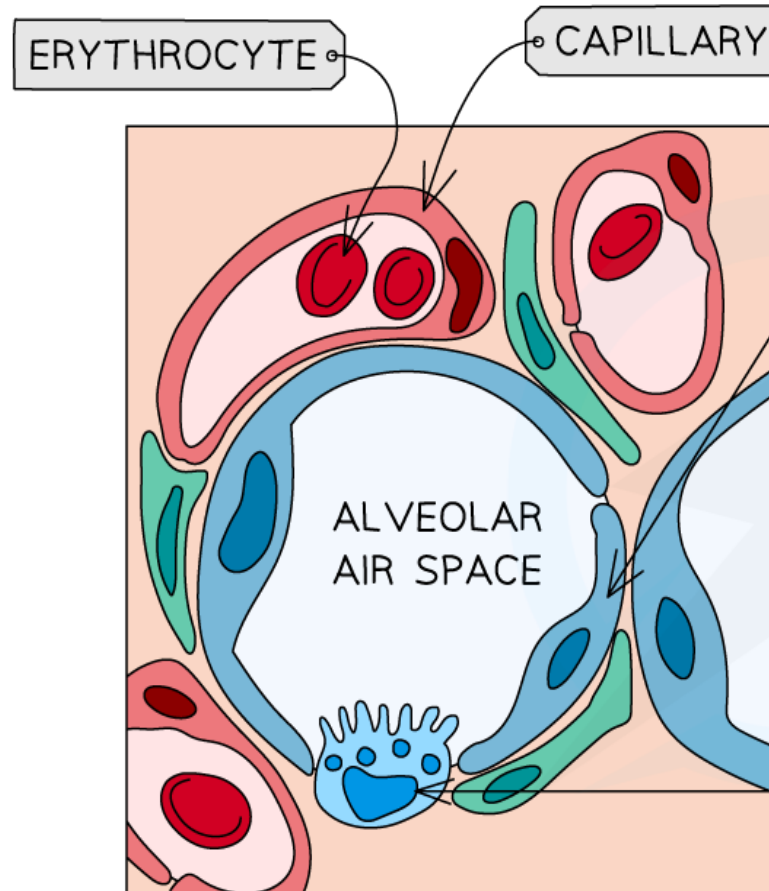


Fig. 15.1 The subdivisions of the airways and their nomenclature. (Modified from Weibel ER: Morphometry of the human lung, Berlin, 1963, Springer.)

Anatomy

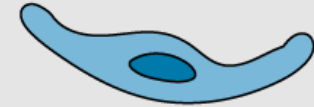
Alveoli:

- 300 million individual sacs
- Type I Pneumocytes (95%)
- Type II Pneumocytes (5%)



TYPE I PNEUMOCYTE

- EXTREMELY THIN CELLS
- CARRY OUT GAS EXCHANGE
- COVER APPROXIMATELY 95% OF THE ALVEOLAR SURFACE AREA

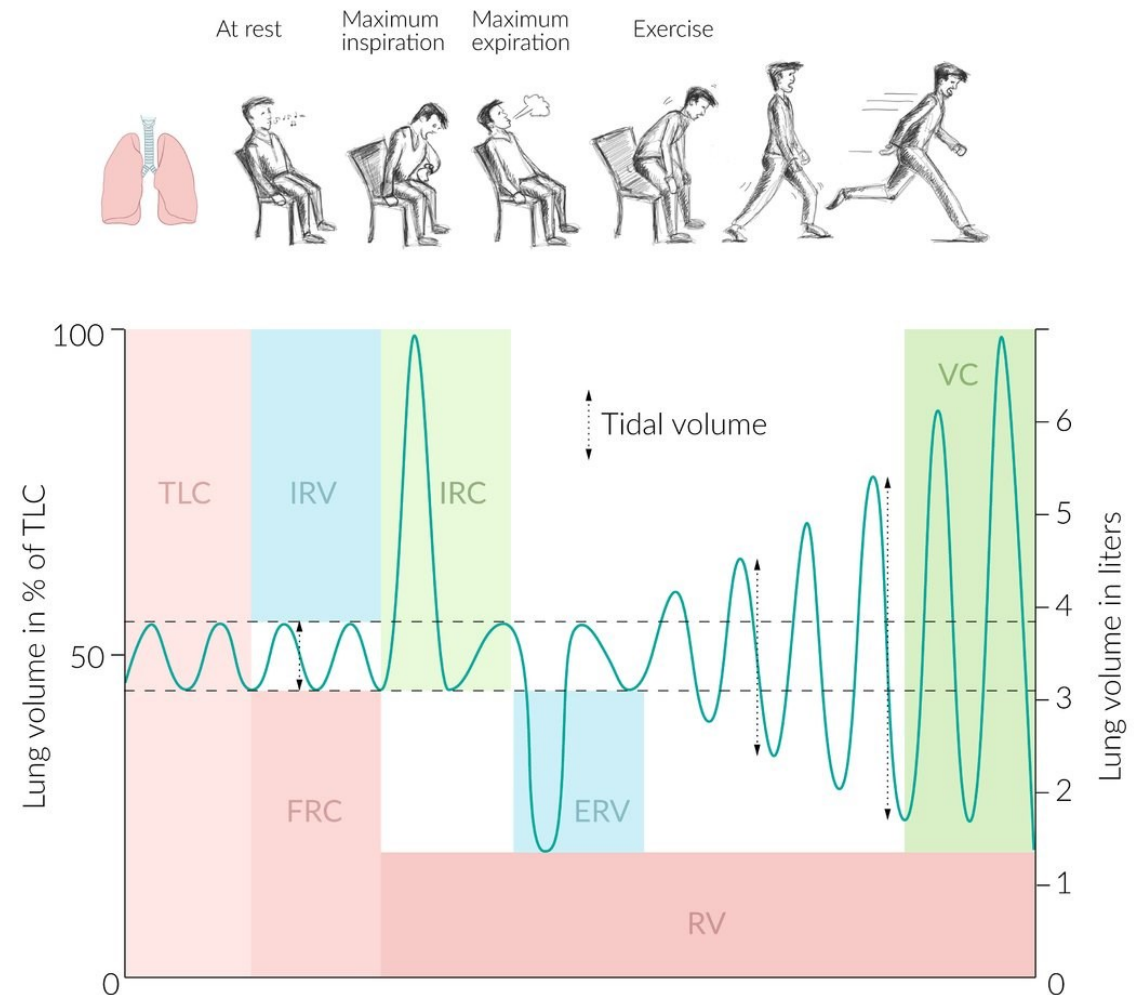
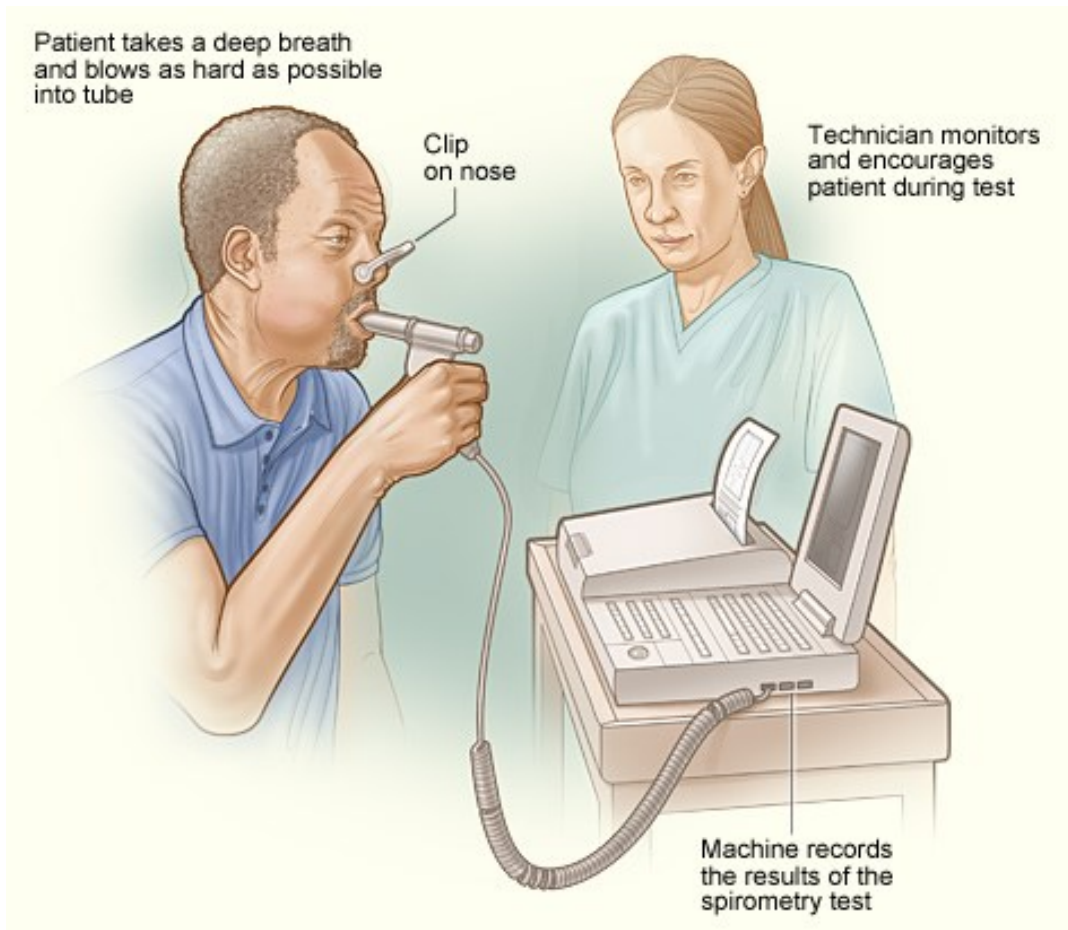


TYPE II PNEUMOCYTE

- ROUNDED CELLS
- SECRETE PULMONARY SURFACTANT
- COVER APPROXIMATELY 5% OF THE ALVEOLAR SURFACE AREA



Physiological lung volumes



Normal lung volumes

	Definition	Normal range
Total lung capacity (TLC)	Volume of air in the lungs after maximal inhalation	6–6.5 L
Vital capacity (VC)	Difference in lung volume between maximal exhalation and maximal inhalation	4.5–5 L
Residual volume (RV)	Volume of air that remains in the lungs after maximal exhalation	1–1.5 L
Tidal volume (TV)	Volume of air that is inhaled and exhaled in a normal breath at rest	~ 500 mL or 7 mL/kg
Functional residual capacity (FRC)	Volume of air that remains in the lungs after the exhalation of a normal TV	2.5–3 L

Investigations in chest case

Pulmonary function testing

Pulmonary function tests (PFTs) measure lung volumes and other metrics of pulmonary function.

They can be used to diagnose ventilatory disorders and differentiate between obstructive and restrictive lung diseases.

The **most common PFT is spirometry**, which requires the patient to breathe forcefully through their mouth into an external device.

This simple and cost-effective test measures both dynamic and static lung volume (with the exception of residual volume and total lung capacity) and airflow rates.

Full-body plethysmography measures both residual volume and total lung capacity in a sealed chamber. **Single-breath diffusing capacity** helps determine if the alveolar membrane is thickened (e.g., in pulmonary fibrosis) or destroyed (e.g., in emphysema), and if the pulmonary vasculature is affected (e.g., in pulmonary hypertension).

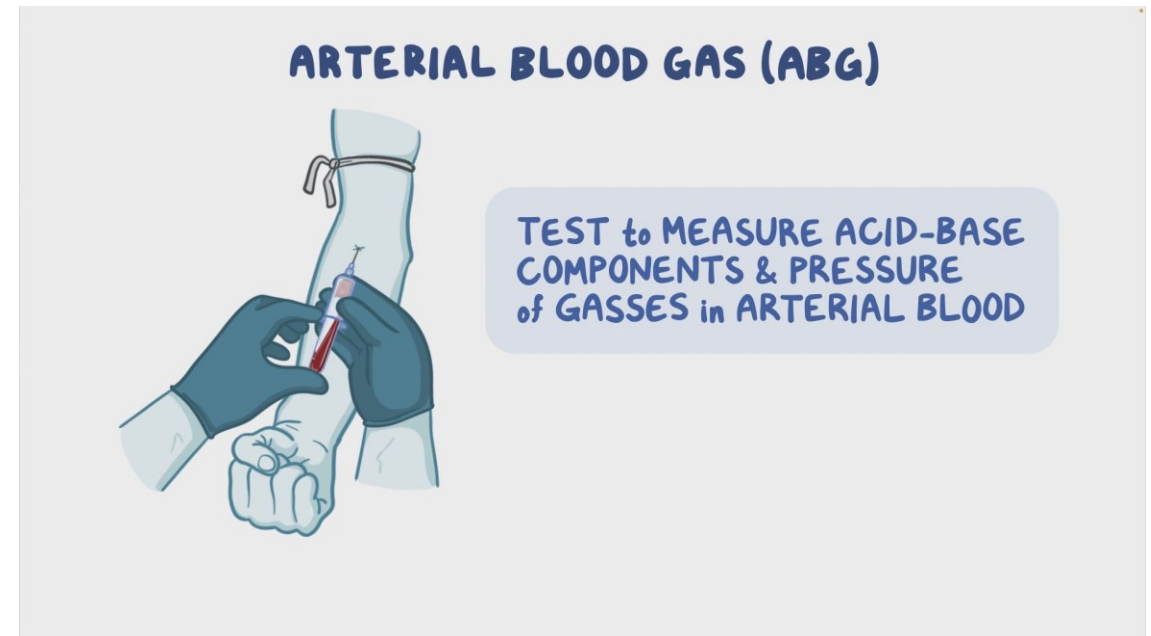
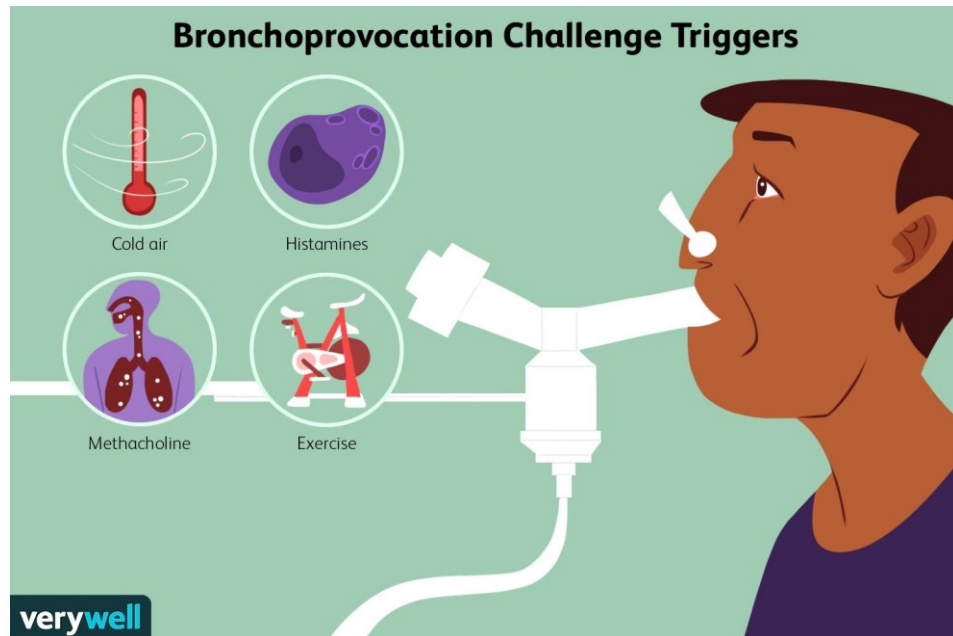
A diagnosis cannot be made based on PFT results alone.

PFT findings in obstructive vs. restrictive lung diseases

		Obstructive lung disease	Restrictive lung disease
Spirometric findings	FEV_1	$\downarrow < 80\%$ of the predicted value or the lower limit of normal	Normal or $\downarrow$
	FEV_1/FVC	$\downarrow$ (because $\downarrow$ in $FEV_1 > \downarrow$ in FVC)	Normal or $\uparrow$ (because $\downarrow$ in FEV_1 is proportional to $\downarrow$ in FVC)
	VC	$\downarrow < 80\%$ of the predicted value or the Lower limit of normal	$\downarrow$
Plethysmograph findings	RV	$\uparrow \geq 120\%$ of the predicted value or the Upper limit of normal	Normal or $\downarrow$
	TLC	Normal or $\uparrow$	$\downarrow$
DL_{CO}		Variable (e.g., $\downarrow$ in emphysema, $\uparrow$ or normal in bronchial asthma, normal in chronic bronchitis)	Variable (e.g., normal with extrinsic causes, $\downarrow$ with intrinsic causes)

Other modalities

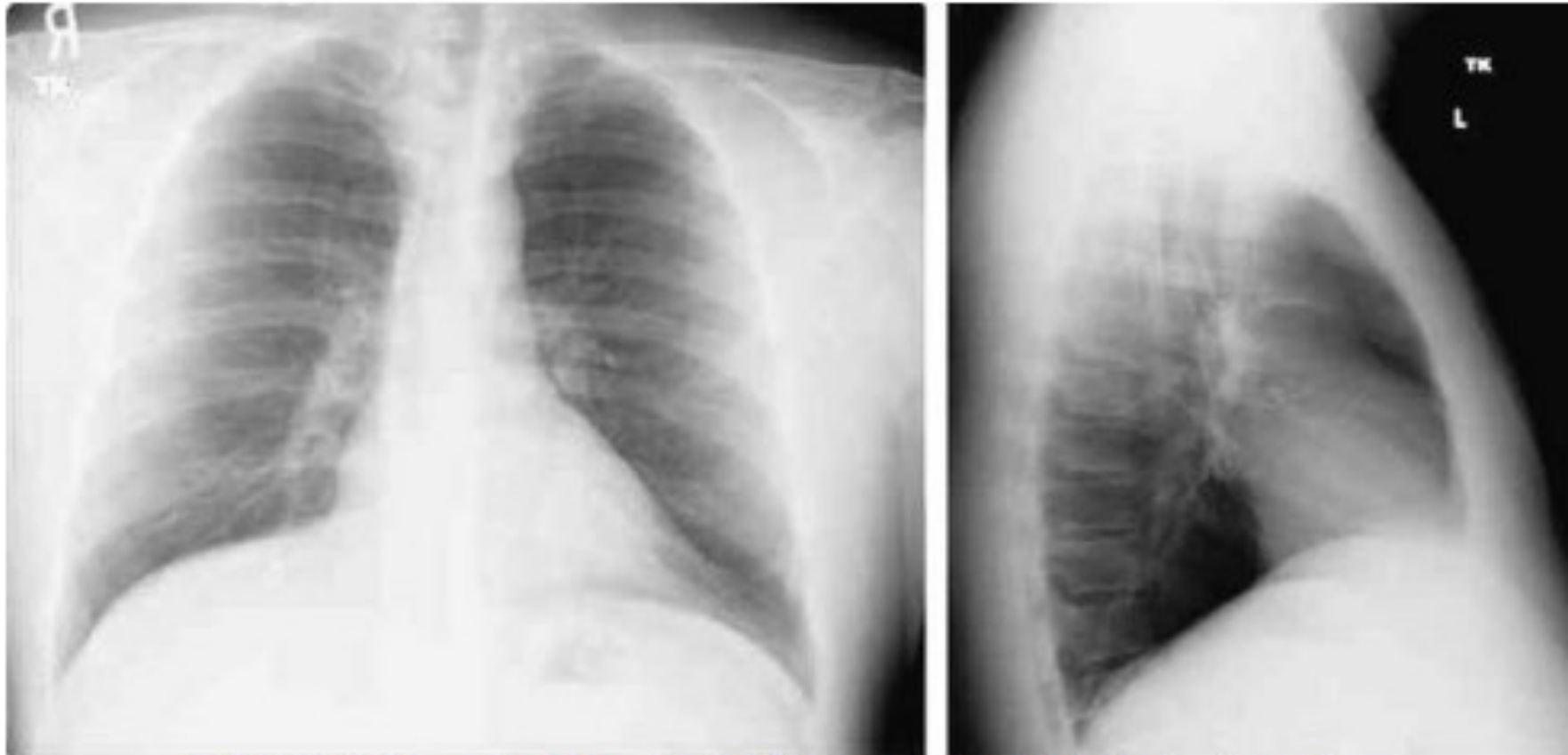
- **Bronchoprovocation Testing**
- **Arterial blood gases (ABG)**
- **6-minute walk distance**



Evaluation of the lung disease

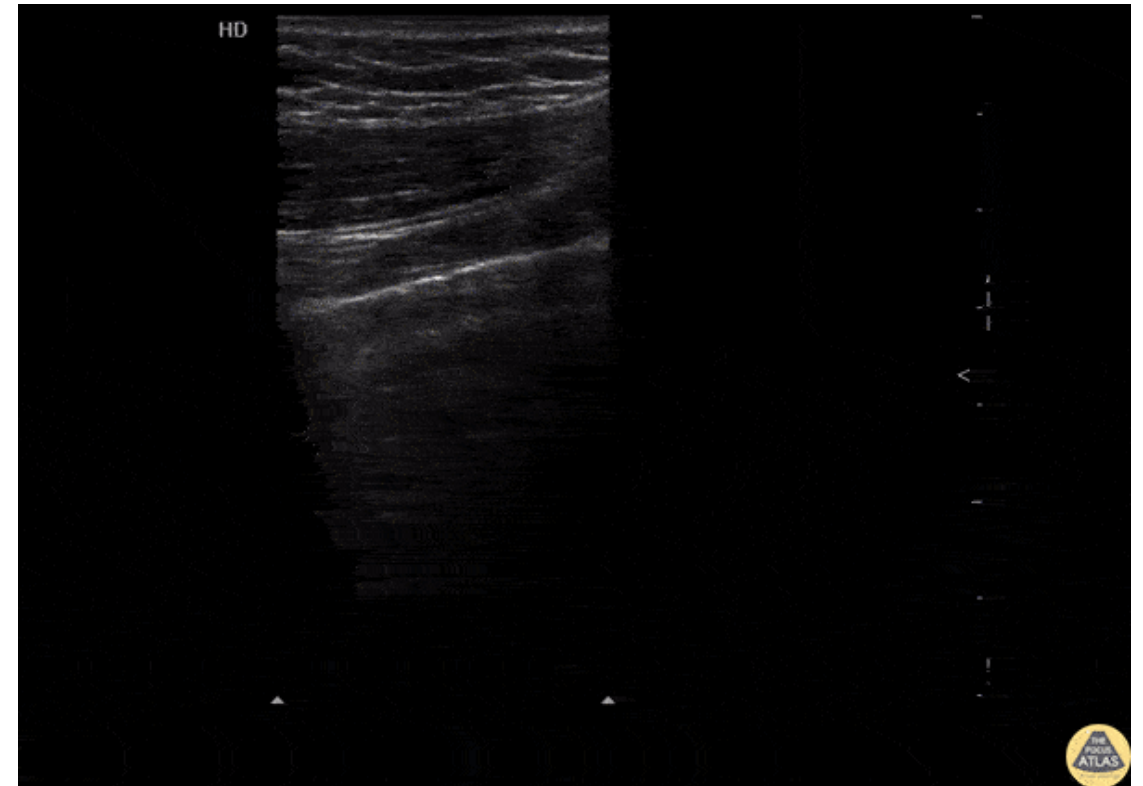
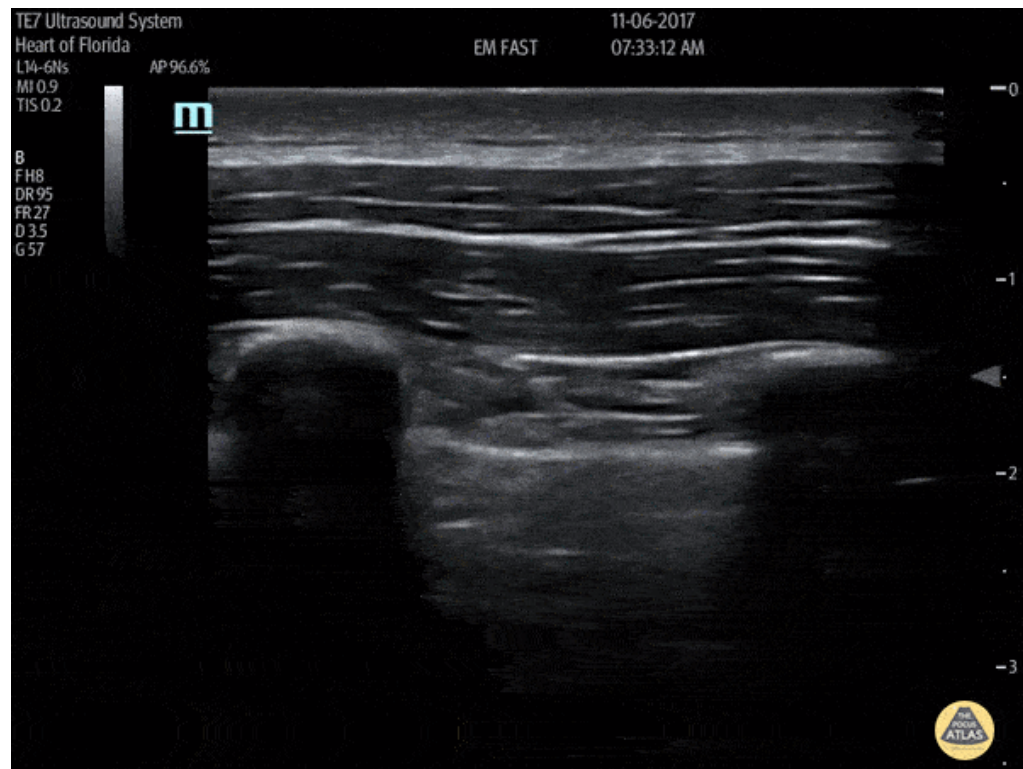
- **Chest Radiography (CXR, CT, MRI, CTPA)**
- **Fluroscopy**
- **Ultrasound (POCUS)**
- **Pulmonary angiography**
- **Pulmonary ventilation perfusion scan (V/Q scan)**
- **Bronchoscopy**

CXR

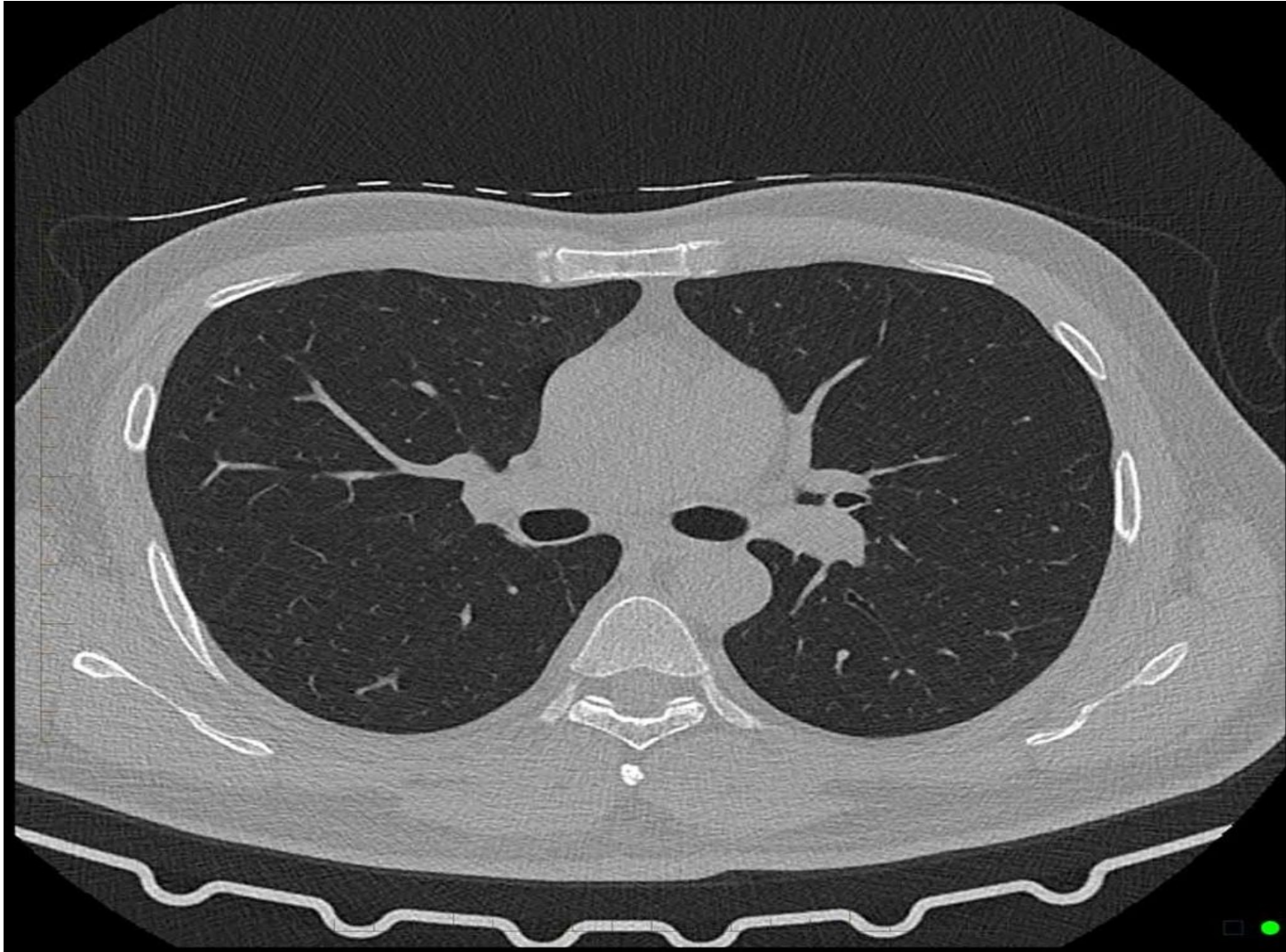


E-Fig. 15.5 Normal posteroanterior (*left*) and lateral (*right*) chest radiographs.

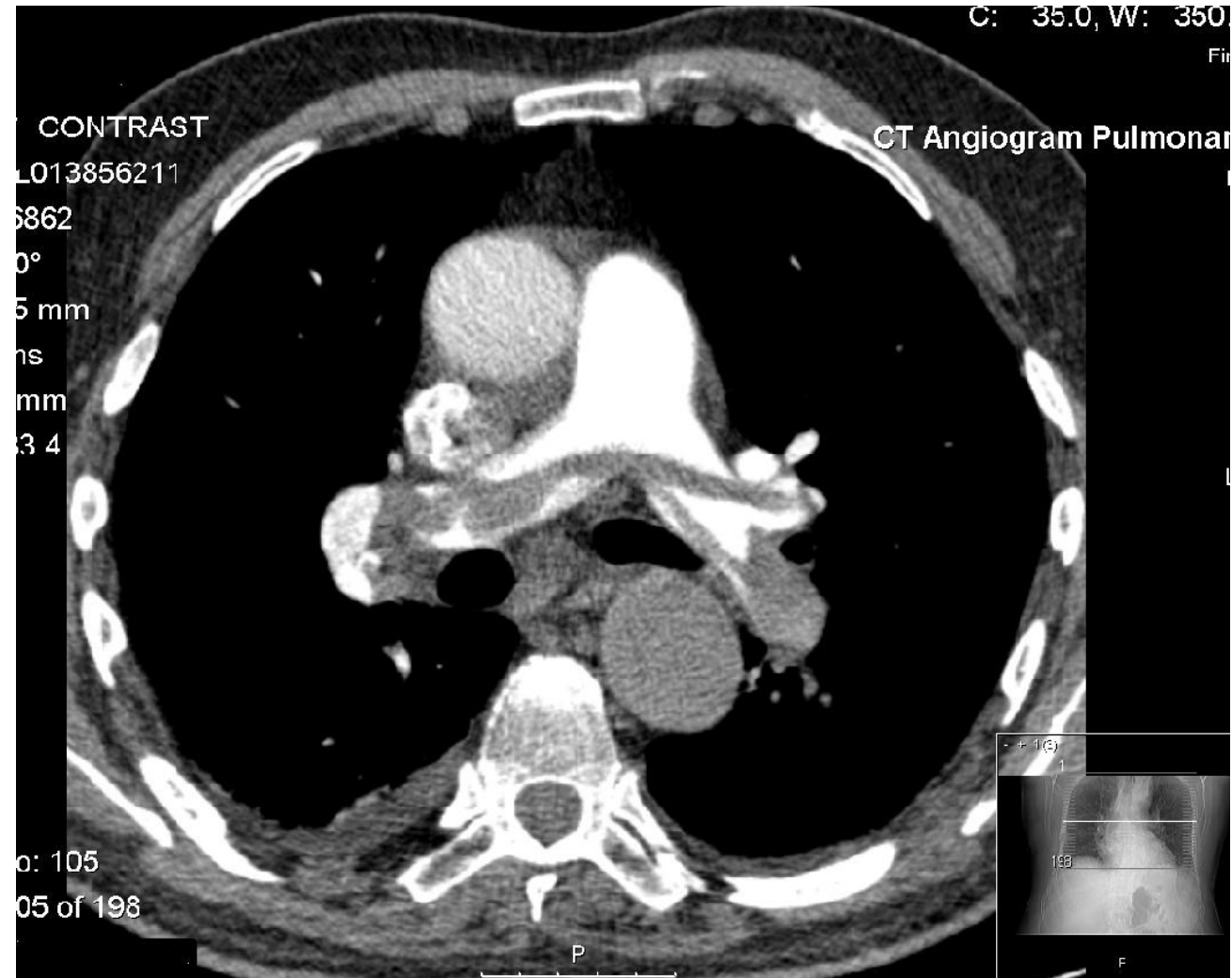
POCUS



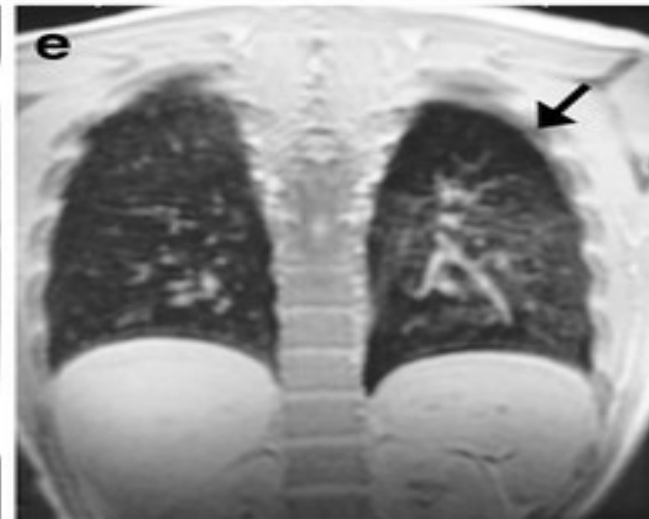
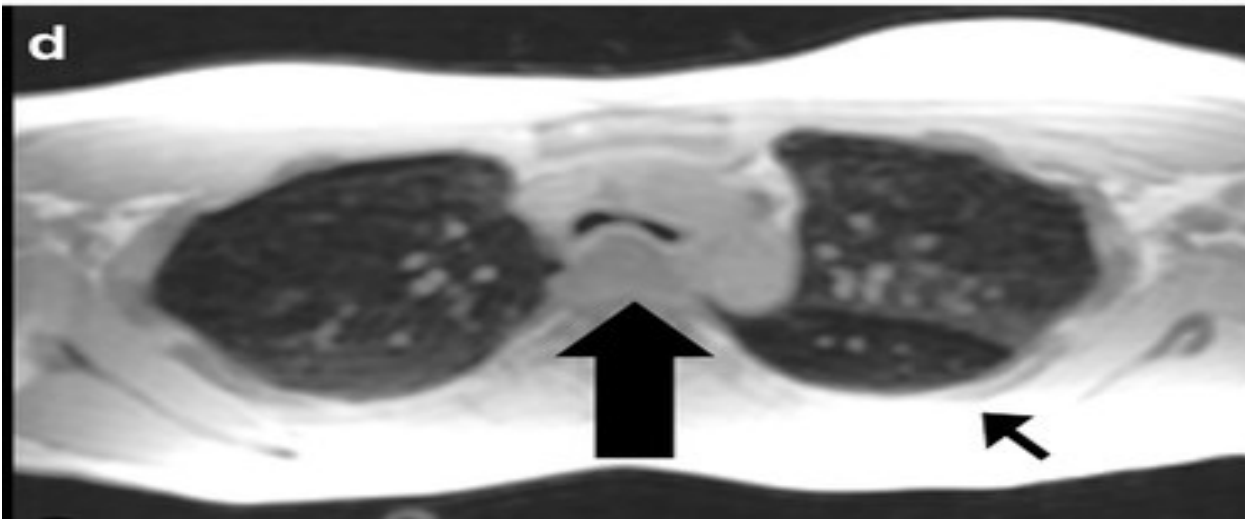
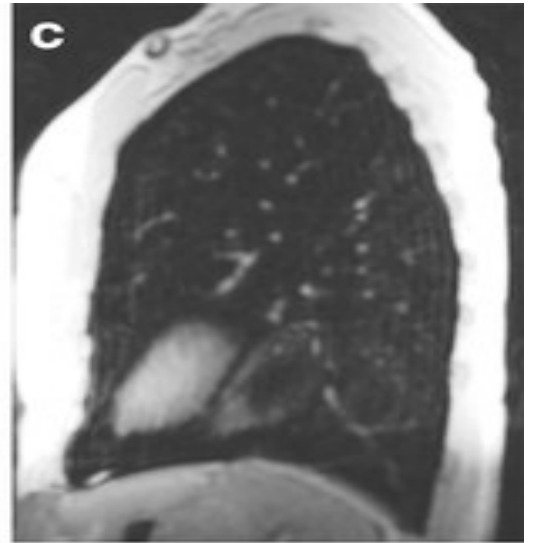
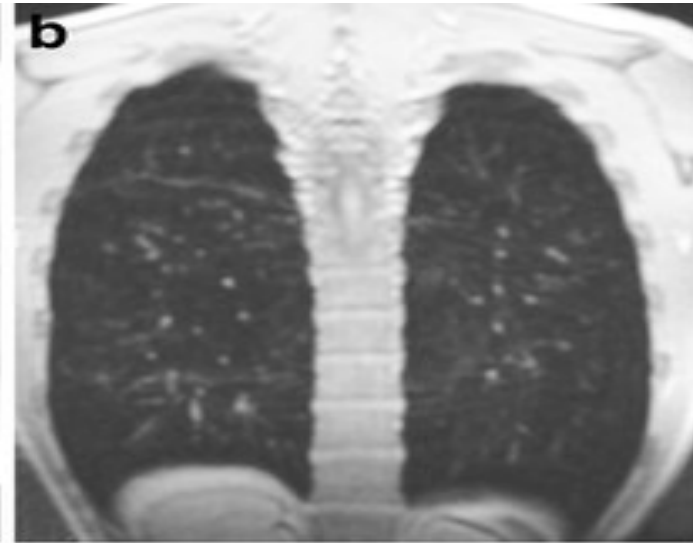
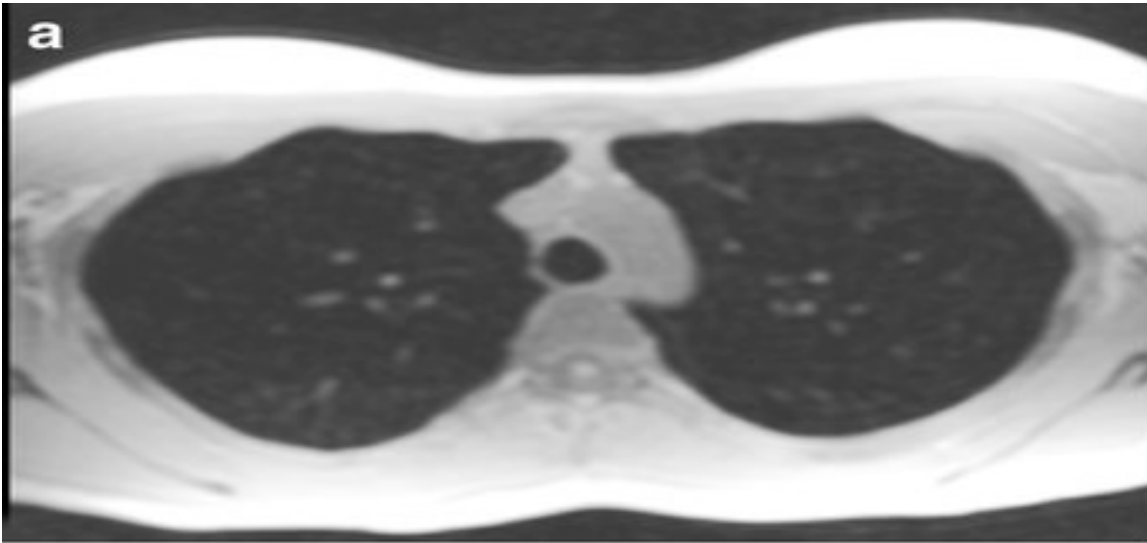
CT Chest



CTPA

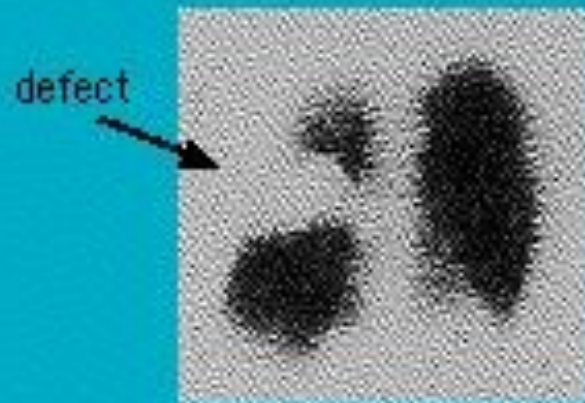


MRI



V/Q Scan

Ventilation/perfusion scan showing
pulmonary embolism in the right upper zone

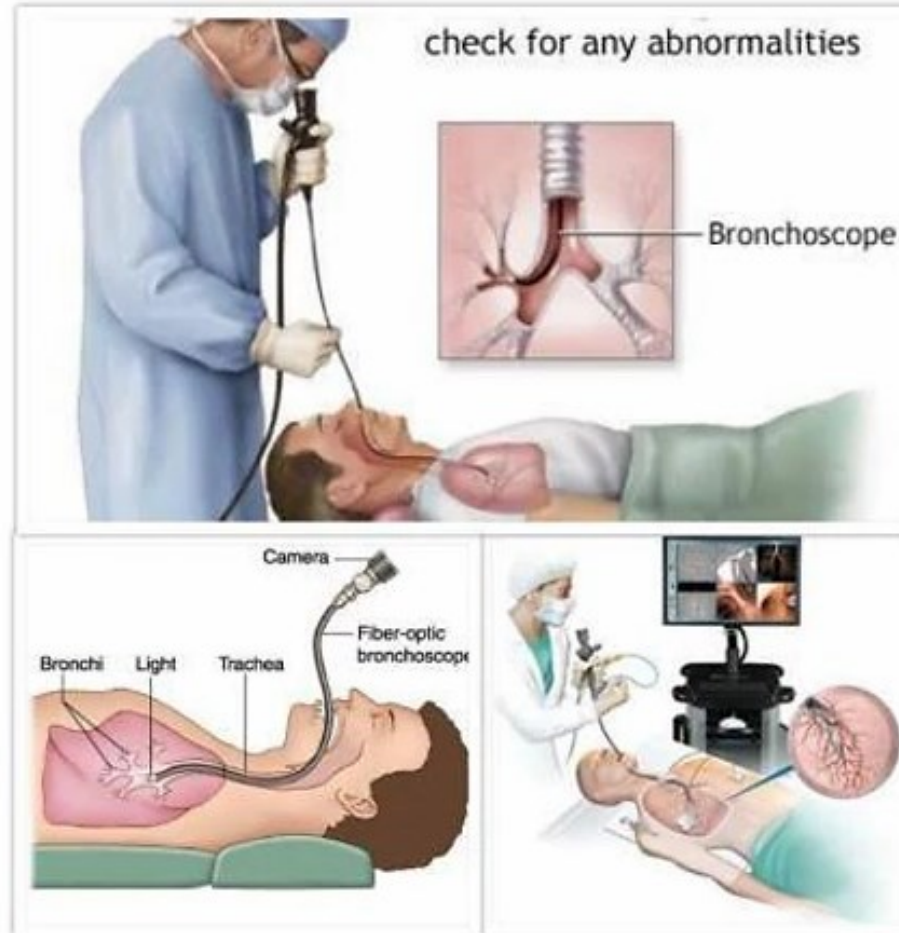


Perfusion scan

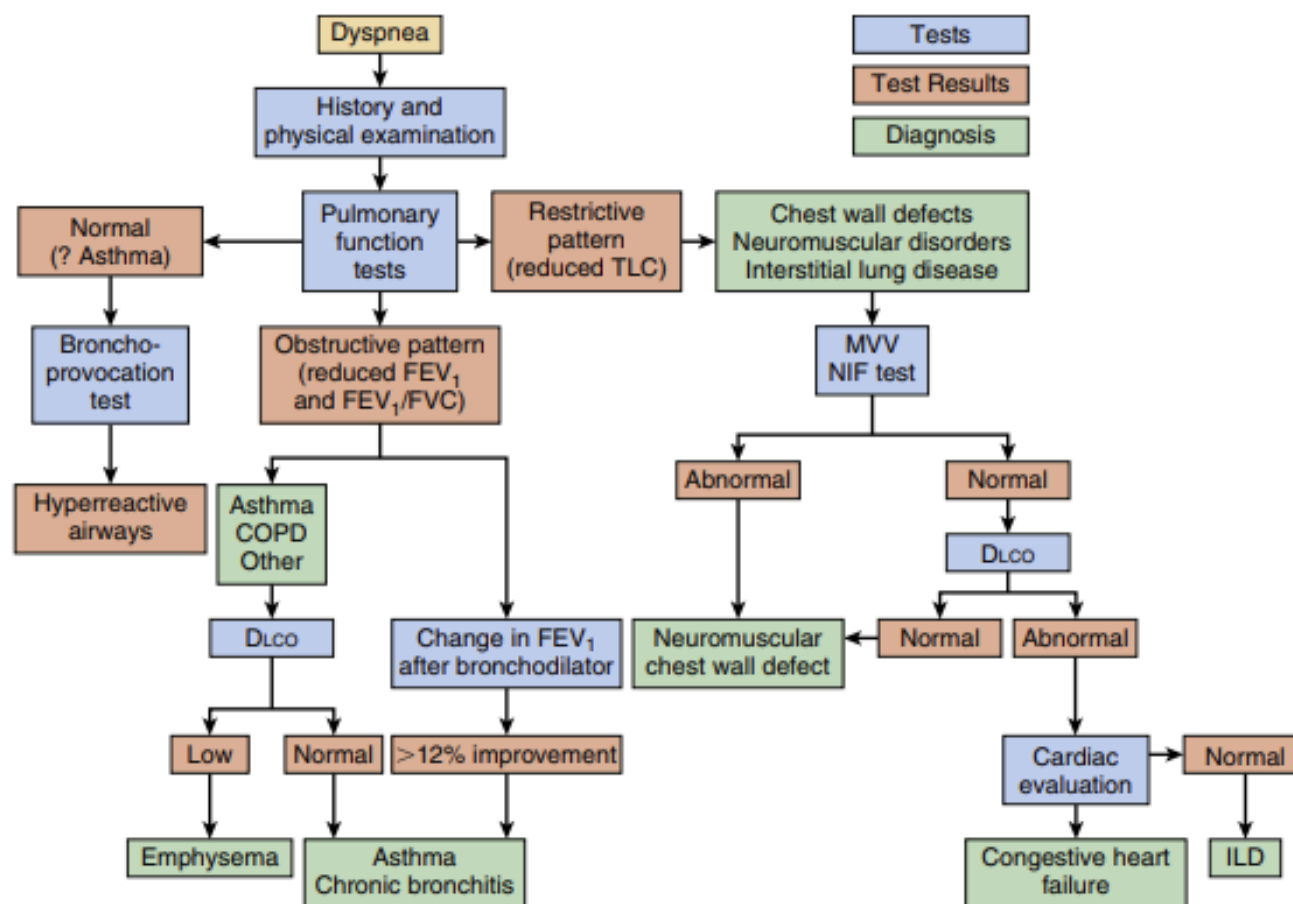


Ventilation scan

Bronchoscopy



To Summarize



E-Fig. 15.4 The flow diagram shows how pulmonary function tests can be used to evaluate patients with obstructive and restrictive disorders. *COPD*, Chronic obstructive pulmonary disease; *DLco*, diffusing capacity for carbon monoxide; *FEV₁*, forced expiratory volume in 1 second; *FVC*, forced vital capacity; *ILD*, interstitial lung disease; *MVV*, maximum voluntary ventilation; *NIF*, negative inspiratory force; *TLC*, total lung capacity.

Case 1

A previously healthy 60-year-old man comes to the physician because of progressively worsening shortness of breath for the past 2 months. He does not experience shortness of breath at rest. He also occasionally has a dry cough. He has not had fever, chills, or night sweats. He has smoked one pack of cigarettes daily for the past 40 years. He drinks a beer daily and occasionally more on weekends. He does not use illicit drugs. He is 183 cm (6 ft 0 in) tall and weighs 66 kg (145 lbs); BMI is 19 kg/m². His temperature is 37.0°C (98.6°F), pulse is 94/min, respirations are 21/min, and blood pressure is 136/88 mm Hg. Lung auscultation reveals a prolonged expiratory phase and end-expiratory wheezing. Spirometry shows an FEV₁:FVC ratio of 62% after bronchodilator inhalation, an FEV₁ of 60% of predicted, and a total lung capacity of 125% of predicted. The diffusion capacity of the lung (DL_{CO}) is decreased. Which of the following is the most likely diagnosis?

- a. Interstitial lung disease
- b. Hypersensitivity pneumonitis
- c. Chronic obstructive pulmonary disease
- d. Bronchiectasis
- e. Bronchial asthma

Questions?

References & further reading

- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- Cecil Essentials of Medicine 10th edition
- POCUS Atlas

Thank You